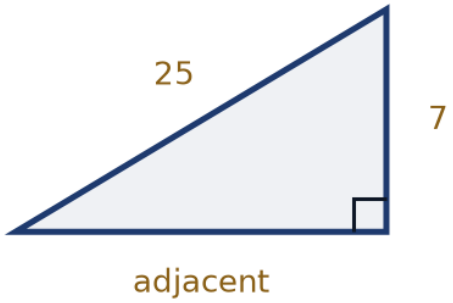
# Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Right-triangle trigonometry

In a right triangle, the angle $\theta$ has an opposite side of 7 and hypotenuse 25. Find $\cos\theta$ and $\tan\theta$.




**2** [Foundational] • Target: Right-triangle trigonometry

In a right triangle, angle $\theta$ has an adjacent side of 8 and opposite side 15. Find $\sin\theta$ and the hypotenuse.

**3** [Foundational] • Target: Exact values and the unit circle

State the exact value of $\sin\left(\frac{\pi}{3}\right)$ and $\cos(45^\circ)$.

**4** [Foundational] • Target: Graphs of trigonometric functions

State the amplitude and period of $y = 4\sin(2x)$.

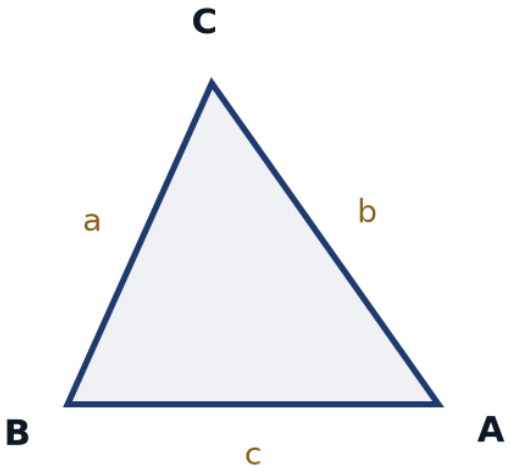
**5** [Foundational] • Target: Trigonometric identities

Given $\sin\theta = \frac{3}{5}$ with $\theta$ acute, find $\cos\theta$ using the identity $\sin^2\theta + \cos^2\theta = 1$.

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**6** [Foundational] • Target: Law of Sines and Law of Cosines

In triangle $ABC$, $a = 9$, $b = 12$, and angle $C = 55^\circ$. Find side $c$, to the nearest tenth, using the Law of Cosines.




**7** [Foundational] • Target: Law of Sines and Law of Cosines

In triangle $ABC$, angle $A = 40^\circ$, angle $B = 70^\circ$, and $a = 8$. Find side $b$, to the nearest tenth, using the Law of Sines.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Law of Sines and Law of Cosines

In triangle $ABC$, $a = 7$, $b = 10$, $c = 13$. Find the largest angle, to the nearest degree, using the Law of Cosines.

**9** [Intermediate] • Target: Trigonometric identities for compound angles

Use the identity $\sin(A + B) = \sin A \cos B + \cos A \sin B$ to find the exact value of $\sin(75^\circ)$, using $75^\circ = 45^\circ + 30^\circ$.

**10** [Intermediate] • Target: A well-designed word problem: the height of a lighthouse

This question has two parts. A ship observes a lighthouse at an angle of elevation of $18^\circ$. After sailing 500 meters directly toward the lighthouse, the angle of elevation increases to $30^\circ$. (a) Let $h$ be the lighthouse's height and $d$ the final distance from its base. Write two equations relating $h$ and $d$ using the tangent ratio. (b) Solve the system to find $h$, to the nearest meter.

**11** [Intermediate] • Target: Graphs of trigonometric functions

[Functions: Modulus & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

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**12** **[Intermediate]** • **Target: Trigonometric identities**

[Advanced Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

**13** **[Intermediate]** • **Target: Solving trigonometric equations**

[Functions, Modulus & Quadratic Theory · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input $t$ when $g(t) = 18$.

**14** **[Intermediate]** • **Target: Law of Sines and Law of Cosines**

[Trigonometry & Circular Measure · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input $t$ when $g(t) = 18$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Trigonometric identities for compound angles** • **Cross-Lesson: 19.01 Normal Distribution**

[Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

**16** **[Challenging]** • **Target: A well-designed word problem: the height of a lighthouse** • **Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

**17** **[Challenging]** • **Target: Right-triangle trigonometry** • **Cross-Lesson: 3.01 Graphs & Transformations**

Given $f(x) = 2x - 3$ for $x \in \mathbb{R}$, sketch $y = |f(x)|$ and solve $|2x - 3| = 5$.

**18** **[Challenging]** • **Target: Exact values and the unit circle** • **Cross-Lesson: 4.01 Coordinate Geometry**

Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta + \alpha)$ where $R > 0$ and $0 < \alpha < 90°$.

**19** **[Challenging]** • **Target: Graphs of trigonometric functions** • **Cross-Lesson: 6.01 Differentiation & Integration**

Show that $x^3 + 2x - 6 = 0$ has a root in $[1, 2]$. Taking $x_0 = 1.5$, perform one Newton-Raphson iteration.

**20** **[Challenging]** • **Target: Trigonometric identities** • **Cross-Lesson: 7.01 Statistics & Probability**

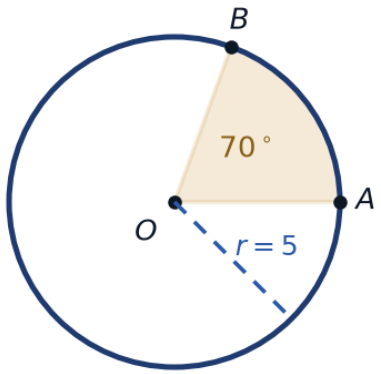
Express $\frac{3x + 1}{(x + 1)(x - 2)}$ in partial fractions.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Right-triangle trigonometry

On the unit circle, a point $P$ is located at coordinates $(x, y)$. If the angle formed by the positive x-axis and the line segment from the origin to $P$ is $\theta$, what is the value of $x$ in terms of $\theta$?




**a.** $\cos\theta$

**b.** $\sin\theta$

**c.** $\tan\theta$

**d.** $1$

**22** [Foundational] • Target: Exact values and the unit circle

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let $\alpha$ be an angle in the first quadrant and $\beta$ be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle $\theta$ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

**a.** $\frac{7}{25}$

**b.** $-\frac{7}{25}$

**c.** $\frac{24}{25}$

**d.** $-\frac{24}{25}$

**23** [Foundational] • Target: Graphs of trigonometric functions

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is $\theta$. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of $P$ is $y = \frac{3}{5}$, determine the x-coordinate of $P$.

**a.** $-\frac{4}{5}$

**b.** $\frac{4}{5}$

**c.** $-\frac{3}{5}$

**d.** $-\frac{25}{16}$

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**24** [Foundational] • Target: Trigonometric identities

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let $\theta$ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

**a.** $-\frac{3}{5}$

**b.** $\frac{3}{5}$

**c.** $-\frac{4}{5}$

**d.** $\frac{4}{5}$

**25** [Foundational] • Target: Solving trigonometric equations

Consider a point $P$ on the unit circle in standard position. If the angle $\theta$ in standard position has a terminal side passing through $P$ such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

**a.** $\frac{\sqrt{3}}{2}$

**b.** $\frac{1}{2}$

**c.** $-\frac{\sqrt{3}}{2}$

**d.** $-1$

**26** [Foundational] • Target: Law of Sines and Law of Cosines

On the unit circle, a point $P(x, y)$ corresponds to an angle $\theta$ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

**a.** $\frac{1}{2}$

**b.** $\frac{\sqrt{2}}{2}$

**c.** $\frac{\sqrt{3}}{2}$

**d.** 1

**27** [Foundational] • Target: Trigonometric identities for compound angles

In triangle $ABC$, the lengths of sides $a$, $b$, and $c$ are opposite to angles $A$, $B$, and $C$ respectively. Given that $a = 7$, $b = 10$, and angle $C = 60^\circ$, find the exact length of side $c$.

**a.** $\sqrt{79}$

**b.** 9

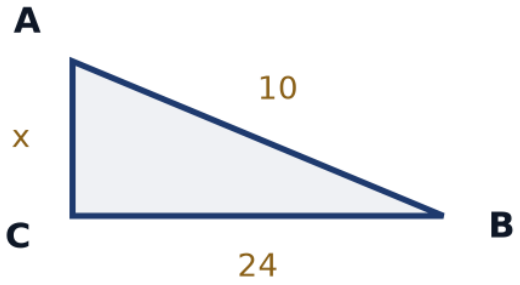
**c.** $\sqrt{91}$

**d.** 10

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**28** [Foundational] • Target: A well-designed word problem: the height of a lighthouse

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse $AB$.




**a.** 25

**b.** 26

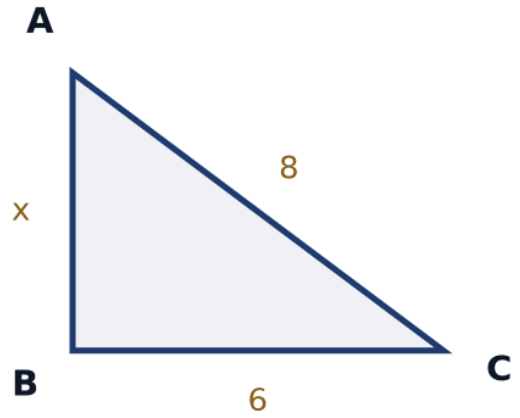
**c.** 27

**d.** 28

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Right-triangle trigonometry

In a right-angled triangle $ABC$, the angle $B$ is 90 degrees. If the length of side $AB$ is 8 units and the length of side $BC$ is 6 units, what is the length of the hypotenuse $AC$?




**a.** 10 units

**b.** 14 units

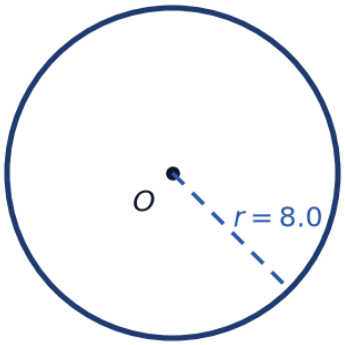
**c.** 48 units

**d.** 20 units

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**30** [Intermediate] • Target: Exact values and the unit circle

A sector of a circle of radius $r = 8$ cm has central angle $\theta = 0.9$ rad. What is the area of the sector?




**a.** $28.8\ \text{cm}^2$

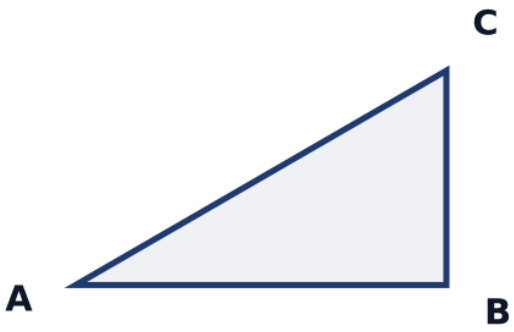
**b.** $57.6\ \text{cm}^2$

**c.** $33.8\ \text{cm}^2$

**d.** $7.2\ \text{cm}^2$

**31** [Intermediate] • Target: Graphs of trigonometric functions

In $\Delta ABC$, $\angle A = 30^\circ$, $\angle B = 90^\circ$, and side $a = 10$. Find side $b$ (Law of Sines).




**a.** 10

**b.** 22

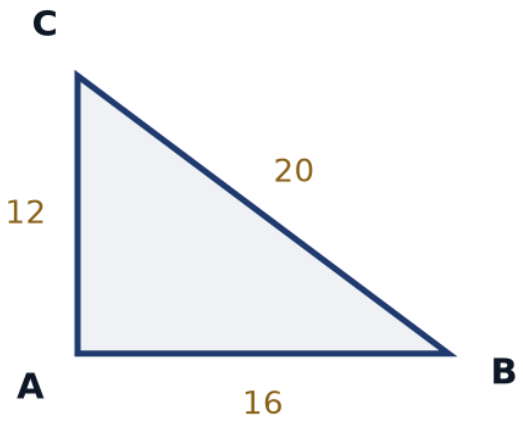
**c.** 5

**d.** 20

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**32** [Intermediate] • Target: Trigonometric identities

A right triangle has legs 12 and 16 and hypotenuse 20. The angle $\theta$ is opposite the side of length 12. Find $\cos(\theta)$.




**a.** $\frac{4}{3}$

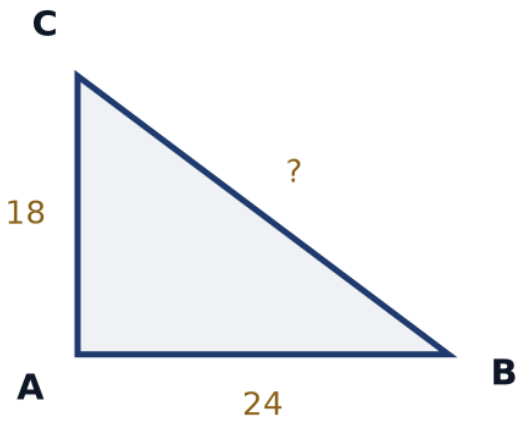
**b.** $\frac{3}{5}$

**c.** $\frac{3}{4}$

**d.** $\frac{4}{5}$

**33** [Intermediate] • Target: Solving trigonometric equations

A right triangle has legs of length 18 and 24. Find the length of the hypotenuse.




**a.** 32

**b.** 34

**c.** 33

**d.** 30

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**34** [Intermediate] • Target: Law of Sines and Law of Cosines

The function $f$ is defined on the open interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by $f(x) = \frac{2-\sin x}{\cos x} + \frac{\cos x}{2-\sin x}$. Determine the exact minimum value of $f(x)$.

**a.** $\frac{4\sqrt{3}}{3}$

**b.** $2$

**c.** $\frac{5}{2}$

**d.** $\frac{3\sqrt{3}}{2}$

**35** [Intermediate] • Target: Trigonometric identities for compound angles

In triangle $ABC$, the side lengths are $AB = (x+2)$ cm, $BC = (3x-1)$ cm, and $AC = 7$ cm. Given that $\angle ABC = 60^\circ$ and $x > 0$, find the exact area of triangle $ABC$.

**a.** $10\sqrt{3}$ cm$^2$

**b.** $5\sqrt{3}$ cm$^2$

**c.** $20\sqrt{3}$ cm$^2$

**d.** $10$ cm$^2$

**36** [Intermediate] • Target: A well-designed word problem: the height of a lighthouse

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If $z$ is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of $r$ and $\theta$.

**a.** $r = 1, \theta = \frac{5\pi}{3}$

**b.** $r = 1, \theta = \frac{4\pi}{3}$

**c.** $r = 10, \theta = \frac{5\pi}{3}$

**d.** $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

**37** [Intermediate] • Target: Right-triangle trigonometry

In triangle $ABC$, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle $ABC$ is $15\sqrt{3}$ cm$^2$. Given that angle $BAC$ is obtuse, find the exact length of $BC$.

**a.** $14$ cm

**b.** $2\sqrt{19}$ cm

**c.** $2\sqrt{34}$ cm

**d.** $8$ cm

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Exact values and the unit circle • Cross-Lesson: 4.01 Coordinate Geometry

A sector of a circle of radius $12$ cm has a central angle of $\frac{\pi}{3}$ radians. Find the perimeter of the sector, giving your answer in exact form.

**a.** $(24 + 4\pi)$ cm

**b.** $(12 + 4\pi)$ cm

**c.** $4\pi$ cm

**d.** $(24 + 24\pi)$ cm

**39** [Challenging] • Target: Graphs of trigonometric functions • Cross-Lesson: 6.01 Differentiation & Integration

In the given right-angled triangle ABC, angle B is 90 degrees. Point D is on BC such that BD = 4 and DC = 9. If AD is the altitude to the hypotenuse AC, find the length of AD.

**a.** 6

**b.** 8

**c.** 12

**d.** 15

**40** [Challenging] • Target: Trigonometric identities • Cross-Lesson: 7.01 Statistics & Probability

In the right-angled triangle ABC, shown below, AD is the altitude to the hypotenuse BC. If BD = 4 and DC = 9, find the length of the altitude AD.

**a.** 5

**b.** 6

**c.** 7

**d.** 8

**41** [Challenging] • Target: Solving trigonometric equations • Cross-Lesson: 8.01 Functions: Composite & Inverse

In $\Delta ABC$, $a = 8$, $b = 10$, $c = 12$. Find $\angle A$ (to one decimal place) using the Law of Cosines.

**a.** $41.4^\circ$

**b.** $46.4^\circ$

**c.** $138.6^\circ$

**d.** $36.4^\circ$

**42** [Challenging] • Target: Law of Sines and Law of Cosines • Cross-Lesson: 9.01 Circular Measure

In $\Delta ABC$, $a = 5$, $b = 6$, $c = 7$. Find $\angle A$ (to one decimal place) using the Law of Cosines. (trial 2)

**a.** $49.4^\circ$

**b.** $135.6^\circ$

**c.** $44.4^\circ$

**d.** $39.4^\circ$

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**43** [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 1.01 Indices, Surds & Quadratics

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

**a.** $\frac{1}{4}$

**b.** $\frac{1}{2}$

**c.** $\frac{3}{5}$

**d.** $-\frac{1}{2}$

**44** [Challenging] • Target: A well-designed word problem: the height of a lighthouse • Cross-Lesson: 10.01 Series & Binomial Theorem

In $\Delta ABC$, $b = 10$, $c = 7$, and $\angle A = 120^\circ$. Find side $a$ (to one decimal place) using the Law of Cosines.

**a.** 13.8

**b.** 14.8

**c.** 15.8

**d.** 12.2

**45** [Challenging] • Target: Right-triangle trigonometry • Cross-Lesson: 11.01 Kinematics

In triangle $ABC$, $a = 5$, $b = 10$, and $C = 120^\circ$. Find the exact length of $c$.

**a.** $5\sqrt{7}$

**b.** $\sqrt{179}$

**c.** 15

**d.** $3\sqrt{19}$

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